

Questions

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Q1 - 2024 (04 Apr Shift 1)

The resistances of the platinum wire of a platinum resistance thermometer at the ice point and steam point are 8Ω and 10Ω respectively. After inserting in a hot bath of temperature 400°C , the resistance of platinum wire is :

(1) 10Ω (2) 8Ω (3) 16Ω (4) 2Ω

Q2 - 2024 (04 Apr Shift 1)

On celcius scale the temperature of body increases by 40°C . The increase in temperature on Fahrenheit scale is :

(1) 68°F (2) 75°F (3) 72°F (4) 70°F

Q3 - 2024 (09 Apr Shift 2)

At room temperature (27°C), the resistance of a heating element is 50Ω . The temperature coefficient of the material is $2.4 \times 10^{-4}^\circ\text{C}^{-1}$. The temperature of the element, when its resistance is 62Ω , is _____ $^\circ\text{C}$.

Questions

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Answer Key

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Solutions

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Q1

Given $R_0 = 8\Omega$, $R_{100} = 10\Omega$

$$\therefore R_{100} = R_0(1 + \alpha\Delta T)$$

Also, $R_{400} = R_0(1 + \alpha\Delta T^1)$

$$\therefore 10 = 8(1 + \alpha \times 100) \Rightarrow 100\alpha = \frac{1}{4}$$

$$\therefore R_{400} = 8(1 + 400\alpha) = 8(1 + 1) = 16\Omega$$

Q2

We know that per $^{\circ}\text{C}$ change is equivalent to 1.8° change in $^{\circ}\text{F}$.

$\therefore 40^{\circ}$ change on celcius scale will corresponds to 72° change on Fahrenheit scale.

Hence option (3)

Q3

$$R = R_0(1 + \alpha\Delta T)$$

$$62 = 50 [1 + 2.4 \times 10^{-4} \Delta T]$$

$$\Delta T = 1000^{\circ}\text{C}$$

$$\Rightarrow T - 27^{\circ} = 1000^{\circ}\text{C}$$

$$T = 1027^{\circ}\text{C}$$

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